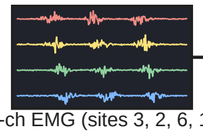
Operator
Module
New (ours)
Supervision
Literal
EMG state
Phoneme token
Disputed token
Trained
Frozen


4-ch EMG (sites 3, 2, 6, 1)

Live hum filter rig/filters.py → 689 Hz

Scale + Session Adapter

Conv Front-end 8× down

Transformer ×12 RoPE · SwiGLU FA2 (H100) / SDPA

86 frames/s EMG states

Letter CTC

Phoneme CTC

Articulator Head

Speaking Head

from parallel audio

SPARC targets ❄

start · end · word ends

to the joint search and the lock

Partial-Sentence Training$^{(F)}$

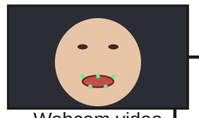
EMG envelope

Rate Token$^{(E)}$ (FiLM)

pace + effort → voice speed, volume

K, V

EMG top-N (letter-CTC beam)


Webcam video

Auto-AVSR re-read every 100 ms ❄

Camera CTC probs per frame: "grab the bat" first

Look-alike Expansion$^{(C)}$ in the joint search

candidate phonemes

Disputed-Span Verifier$^{(B)}$

S_{emg} = CTC + verifier

r: camera conf · face EMG quality · rate

Reliability Gate$^{(D)}$

g_{cam}, g_{emg}, g_{lm}

Joint Search $\Sigma_m g_m s_m$ + hints

Confidence Lock$^{(G)}$

"grab the mat" → Cartesia, call

Homophene-Contrastive Loss$^{(A)}$

Claude predict + repair ❄

whole conversation, judge via Ink

Conversational LM$^{(H)}$ + cache$^{(I)}$

judge (Ink) + locked words

Mouth activity + expression

Endpoint$^{(G)}$: EMG speaking + lips still

camera log-probability

mouth activity, face visible

(A) Homophene-Contrastive Training

true: "grab the mat"

Phonemes → Visemes lexicon + viseme map

Swap Lip Look-alikes m ↔ b ↔ p, f ↔ v, t ↔ d ↔ n

grab the bat | grab the pat | crab the mat

N look-alike negatives

CTC + Verifier Scores for true + N negatives

Softmax over N+1

Cross-Entropy (true sentence must win)

(B) Disputed-Span Verifier

candidate phonemes (Q) | EMG states (K, V)

+ Disputed Embedding (differs across candidates)

Cross-Attention Decoder ×2 phonemes attend to EMG frames

Pool over Disputed Tokens

Linear → Match Score

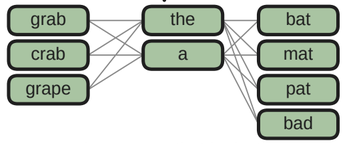
s_{ver} = 2.7

Only the words in dispute decide the score: EMG checks exactly what the camera cannot see.

(C) Look-Alike Expansion

a draft in the beam: "grab the bat"

Phonemes → Visemes



every word sequence that looks the same

LM$^{(H,I)}$ + Claude hints prune

EMG Verifier picks$^{(B)}$

"grab the mat"

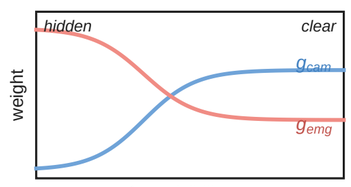
(D) Reliability Gate

camera confidence | face visible | EMG quality | speaking rate

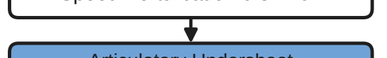
MLP

Softmax

g_{cam}, g_{emg}, g_{lm}



(E) Fast-Speech Robustness


normal pace | fast: smaller, overlapping

Speed Perturbation 0.8–1.6×

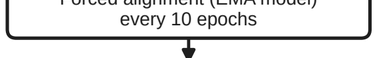
Articulatory Undershoot peak compression + burst overlap

Envelope → Rate Estimate

FiLM: γ(rate) $h + \beta$(rate) on each block's normalised input

Transformer ×12

(F) Partial-Sentence Training


kept: frames before t | *cut*

Forced alignment (EMA model) every 10 epochs

Cut 30% of rows at a random frame t (30–100% of length)

Target = letters aligned before t "i want wa…"

Letter CTC + speaking loss on the cut

The live loop reads beginnings of sentences every 100 ms, so SVR-1 trains on them.

(G) Real-Time Loop: text and voice while you mouth

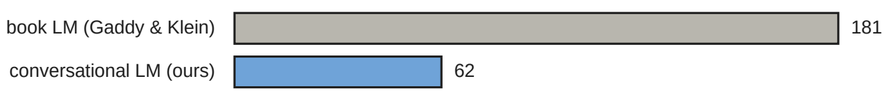
Claude predicts likely words before you mouth → Every 100 ms while mouthing: re-read clip, camera + EMG → Joint search: camera + EMG + LM$^{(H,I)}$ + Claude hints → Confidence lock: one learned score, threshold moves with context → Locked words: grey → black + Cartesia voice into the call

Endpoint$^{(G)}$: ~0.5 s of EMG quiet + lips still → Final pass: camera beam + look-alikes$^{(C)}$ + verifier$^{(B)}$ + Claude → Corrected text replaces live text → Still unsure: "Please repeat" + reasons in the error log

(H) Conversational Language Model

72M words of dialogue subtitles · Cornell films DailyDialog · Topical-Chat → Normalised like SVR-1 recording prompts removed → Pruned word 4-gram (KenLM) 110 MB, loads in 0.4 s → Every hypothesis, every pass ~2 µs per word

Perplexity on our 600 recording prompts (lower is better: the LM expects how people talk)



size 946 MB → 110 MB · load 5.2 s → 0.4 s · words never seen 8 → 3

(I) Call Cache: the conversation so far

Judge's lines (Cartesia Ink)

Austin's locked words

→ Call cache: bigram + unigram counts, fading (half-life 400 words) → $P = (1-\lambda) P_{ngram} + \lambda P_{cache}$ λ: 0 → 0.15 as the call grows → to the joint search (g_{lm})

judge: "hi, this is Priya from admissions" → "i talked to priya …": ln P −33.7 → −25.3 (×4000)

A name just mentioned becomes likely at once; its look-alike "pria" does not. Claude stays separate: hints.